

National Land Acquisition & Management System

Smart India Hackathon (SIH) 2026 Technical Report

MEMBER PROFILE: Member 5

Technical Role: Security, RBAC & Immutable Audit Trail Engineer

Primary Focus: Endpoint authentication scopes and permission checking dependencies, Password cryptography ha

1. Functional Domain & Work Done

As part of the NLAMS development team, Member 5 has taken ownership of the functional module targeting Security, RBAC & Immutable Audit Trail Engineer. The following components have been fully implemented and integrated into the live workspace:

- **XSS Sanitizer Middleware:** Built FastAPI middleware filtering request payloads to strip HTML/Script tags, preventing stored script injection breaches.
- **CORS Access Control Lists:** Restructured FastAPI CORS origin validation to strictly match local ports and Cloudflare tunnel domains.
- **Secure Wallet Checks:** Implemented dependency-based wallet address checks on direct-benefit-transfer escrow releases.

2. Overall Website Layout & Features

The NLAMS portal operates as a single source of truth under a modernised, high-contrast, official government light theme (saffron, navy, and emerald accents). Key features include:

- **Tricolor Home & Directives:** An entry portal showcasing the LARR Act 2013 roadmap and operational pillars.
- **Bilingual Language Switcher:** A global toggle translating all KPI labels, timelines, Gazette notifications, and Treasury advice receipts dynamically into Hindi and English.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons, color-coded by acquisition stage, alongside Recharts predictive dashboards.
- **Automated LARR Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust & Compensation Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile-Responsive Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.

3. Sequential Learning Pathway & Project Roadmap

To ensure a smooth transition from prototype to production for the SIH finals, the following learning benchmarks are mapped for this role:

- **Focus Area:** Endpoint authentication scopes and permission checking dependencies
- **Focus Area:** Password cryptography hashing (bcrypt via passlib)
- **Focus Area:** Immutable append-only database audit logging
- **Focus Area:** XSS inputs sanitizers and HTML strip middleware

4. Judge Defense Recommendations

During the pitch and technical Q&A, the judges will test the integrity of each module. Use the following defense arguments to demonstrate technical authority:

- **Defense Tip:** Explain that database audit logs use append-only constraints with zero UPDATE or DELETE permissions, guaranteeing a cryptographically tamper-proof paper trail.